

Deep Learning Based Cancer Detection in Bone Marrow using Histopathological Images

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Abstract—The cancerous proliferation of abnormal white blood cells (WBC) in the bone marrow and blood is characteristic of leukemia, which has its origins in these blood-forming tissues. Blasts are immature aberrant cells that inhibit the creation of new, lineage-normal blood cells. Acute Lymphoblastic Leukemia (ALL), Acute Myeloid Leukemia (AML), Chronic Lymphocytic Leukemia (CLL), and Chronic Myelomonocytic Leukemia (CML) are the four Chronic Lymphocytic Leukemia (CLL), Chronic myeloid leukemia (CML), and Acute myeloid leukemia (AML) (CML). Patients with leukemia are often diagnosed after their symptoms have been evaluated, a complete blood count (CBC) has been run, and a peripheral blood smear has been examined under a microscope by a pathologist. As part of the process to confirm and classify leukemia, a bone marrow examination and other sophisticated laboratory testing are performed. Traditional blood and bone marrow smear examination using light microscopy is plagued by intra- and inter-observer variability. Overcoming these individual biases of doctors is possible with the use of image processing-based approaches that can automatically analyse pictures of blood and bone marrow smears to detect abnormal cells. Using an American Society of Hematology dataset, the researchers classified cases of ALL into B-cell and T-cell subtypes, as defined by the World Health Organization (ASH). The research improves classification accuracy to 94.12% and compares the effectiveness of three distinct training strategies. The research also includes subtyping AML into M3, M4, and M5 using a ResNet-50 pretrained network and traditional machine learning based classifiers based on the CNN's characteristics. Using the features of ResNet-50, we were able to get a classification accuracy of 96.43 percent, and we were able to get a hundred percent accuracy using just a small number of traditional classifiers. Using the properties of Element and an ASH dataset, the researchers also sought to classify cases of chronic leukemia into CLL and CML. Classification using the Element yields an accuracy of 92.59%, while classification using the SVM with the Lucent's deep features yields an accuracy of 98.15%. The study's classifications are based on a straightforward framework that doesn't need sophisticated picture segmentation or feature extraction. The research has various limitations, one of which is that the dataset utilized in some of the classifications is not particularly big, which leads to a loss in classification accuracy. To address this, the research used data augmentation methods.

Keywords—*cnn algorithm, plane extraction, linear distinction stretching, world thresholding, gray level co-occurrence matrix, sickle cell anemia.*

I. INTRODUCTION

Cancer is the name given to a group of diseases that are life-threatening and are characterized by the development of abnormal cells that divide rapidly and uncontrollably [1] and spread to other tissue and organs of the body [2]. Cancer is the name given to the disease that is associated with the development of these abnormal cells. According to the Mayo Clinic in Rochester, cancer ranks as the world's second greatest cause of death [3]. The top place goes to cardiovascular disorders (CVD). There are more than 200 distinct forms of cancer (Cancer Research UK, 2020). The following are some of the primary clinical terminology that are used to depict the many forms of cancers:

A. Carcinoma

It is the most prevalent kind of cancer and is produced by epithelial cells, which are the cells that line the interior and exterior surfaces of many organs throughout the body [4].

Sarcoma is a kind of tumor that develops in connective tissues [5]. These connective tissues include bone, cartilage, muscle, fat, blood arteries, and lymph vessels, among other things.

According to the National Cancer Institute in the United States of America [6], leukemia is a non-tumor type of cancer that begins in the blood-forming tissues of the bone marrow and results in a large number of abnormal white blood cells in the blood and bone marrow, which crowds out normal blood cells.

B. Lymphoma

Lymphoma is a kind of blood cancer that originates in the disease-fighting cells known as lymphocytes [7]. Lymphomas are characterized by the accumulation of aberrant lymphocytes in the lymph nodes and other organs throughout the body.

C. Myeloma

It is a kind of blood cancer that originates from the plasma cells, and another name for it is multiple myeloma [8]

According to the World Health Organization's 2020 report, low- and middle-income nations account for around 70% of all cancer fatalities. A high body mass index, a poor intake of vegetables and fruits, a low level of physical

activity, the use of tobacco products, and the use of alcohol are the five primary behavioral and lifestyle factors that may be connected to almost one third of all cancer deaths (WHO, 2020). Tobacco smoking is the single most significant risk factor for cancer, and it is responsible for around 22% of all deaths from the disease [9]. Early detection and treatment of a disease may significantly cut the risk of death caused by that illness, making early detection and treatment of cancer one of the most critical factors in lowering the overall mortality rate caused by cancer.

In 2018, there were 437033 new instances of leukemia reported all over the globe, and the breakdown of the incidence of leukemia cases reported all over the world in 2018 is shown in the form of a chart in Fig. 1.

The specific symptoms of leukemia vary based on the subtype of the disease that a person has. On the other hand, the following are some of the most common signs and symptoms of leukemia. Symptoms such as fever or chills, persistent fatigue, weakness, frequent or severe infections, weight loss, swollen lymph nodes, an enlarged liver or spleen, easy bleeding or bruising, recurrent nosebleeds, tiny red spots on the skin, excessive sweating (especially at night), bone pain or tenderness, etc. are all indications that something may be wrong. Other symptoms include bone pain or tenderness [5]. Fig. 1 shows the sample image.

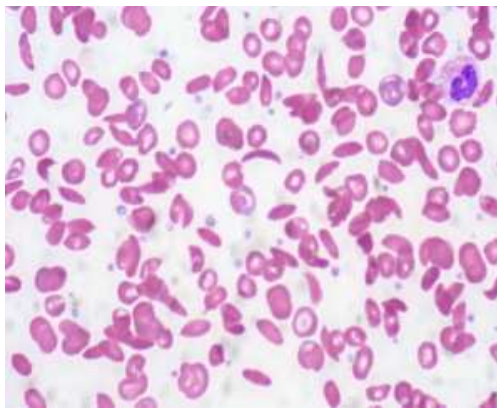


Fig. 1 Sample Image

The examination of smears taken from the patient's blood and bone marrow using a microscope is the primary and traditional method for diagnosing leukemia. More sophisticated laboratory tests are used in order to categorize leukemia into its many forms and subtypes

For an extremely extended period of time, the method of light microscopy has been used for the purpose of seeing both blood and tissue samples. When determining whether a cell is normal or leukemic, hematologists and pathologists who utilize light microscopy rely on the experience and information that they have acquired over the course of their careers. These judgment-based diagnoses, on the other hand, are prone to intra-observer variability, which refers to some variation in results when the same sample is analyzed again at a later time by the same person, and inter-observer variability, which refers to some variation in results when the same sample is analyzed by a different person. In other words, there is some variation in results when the same sample is analyzed by two different people. Processing and analyzing the images of blood and bone marrow smears can be used to solve these problems. These problems can be solved by using image processing-based methods to assist

physicians in the primary diagnosis to detect the presence of leukemic blasts and to classify them. This can be done by processing and analyzing the images of blood and bone marrow smears. These image analysis techniques are not only rapid and inexpensive, but they also do not depend on the competence and ability of the medical experts that are engaged in the diagnosis. In the following paragraphs, we will provide an overview of the image-processing based approaches that may be used for analyzing hematological images.

D. Image Processing-Based Methods

When W. C. Roentgen discovered X rays in 1895, it marked the beginning of a new era in the field of medical imaging, which led to the development of diagnostic imaging techniques. Yet, the first time computers were used in clinical practice and medical imaging was not until the 1970s, when [10] Since that time, computers have become an essential component of the healthcare business, playing a role in a wide number of processes, including the development of images, the presentation of images, and the analysis of images. These days, computers are used in almost every aspect of medical imaging systems, including radiography, ultrasound, magnetic resonance imaging (MRI), and nuclear medicine systems. It was the introduction of digital mammography in the 1990s that paved the way for the widespread use of computer assisted diagnostics (CAD) systems for the early identification and accurate diagnosis of a wide variety of malignancies. The development of image processing and computer vision technologies that allow for quantitative analysis of histopathological and hematological pictures is ultimately made possible as a result of the success of such diagnostics systems. The purpose of computer-aided diagnostic (CAD) systems for the diagnosis of different illnesses is not to take the place of the knowledge and expertise of a physician but rather to help clinicians in making a diagnosis that is simple and efficient.

The role that CAD systems play in hematological image analysis for the diagnosis of leukemia and other diseases primarily involves capturing images of blood and bone marrow smears with camera attached microscopes, performing some kind of preprocessing to remove any noise, performing automated counting of various blood cells, and performing morphological image analysis to detect the presence of abnormal cells. It should be emphasized here that the CAD systems can only assist the physicians in the primary diagnosis of the various types of leukemia, and that the confirmation of the disease is to be done by clinical correlations with the signs and symptoms of the patients and with the assistance of advanced laboratory tests. Experiments have been carried out as part of this proposed work to develop computer-aided diagnosis (CAD) systems for the purpose of assisting medical professionals in the primary diagnosis of leukemia in general, as well as to make some classifications of acute and chronic leukemia for the primary diagnosis of the disease based on the morphological features of blood smear images. These classifications have been made using the morphological characteristics of blood smear images. In comparison to more sophisticated clinical tests and analyses, the analysis that is based on image processing is simple, quick, and more cost-effective. In contrast to traditional procedures, which require delivering blood and bone marrow samples, those based on image processing may perform remote analysis thanks to the ease with which an image can

be sent to any place. This is one of the many appealing aspects of these approaches [11].

E. Problem Statement

Image processing-based computer vision is already being used for the analysis of images of bone marrow and peripheral blood smears to assist pathologists and hematologists in the automated detection and classification of a wide variety of diseases, including leukemia. This helps to speed up the diagnostic process. During the years, a significant number of researchers have focused their efforts on this field to build automated systems capable of detecting a wide variety of illnesses, including malaria, skin disorders, leukemia, and others. There has already been a significant amount of study conducted in the field of automatically detecting leukemia, and the researchers explored a variety of methodologies and analytic methods to identify and categorize leukemia. The majority of the research that fall under this category make use of some kind of picture segmentation, which is then typically followed by some feature extraction techniques and classification using traditional Machine Learning (ML) methods, with the purpose of automating the diagnostic process. The majority of these techniques for picture segmentation and feature extraction entail complicated calculations, which makes it difficult to put them into practice. In addition, there are not many research accessible in the literatures for computer-assisted diagnosis of leukemia utilizing Deep Learning (DL), which is a subfield of machine learning. This is a significant limitation of the field. Hence, there is sufficient room for the development of CAD systems that are straightforward and need a lower level of complexity in order to carry out automated leukemia identification and categorization. In addition, the number of research that have been conducted on the topic of automated chronic leukemia identification and categorization is quite restricted.

II. LITERATURE SURVEY

In order to locate the nuclei of the white blood cells, [12] carried out a few image arithmetic operations as well as a few threshold operations. In the process of designing this system, one of the most difficult tasks was determining an adequate threshold for the segmentation. When it comes to the segmentation of white blood cell nuclei, the threshold that was employed in the system that was designed does not produce an efficient outcome.

[13] created a technique that can identify five different kinds of leucocytes from a picture of blood. At this step of the procedure, nuclei were located initially, and then area growth methods were used to determine the whole of the membrane. The outcomes that were accomplished were quite satisfying.

White blood cell segmentation was accomplished by [14] by the use of certain threshold operations, a low-pass filter for the elimination of background, and clustering. The photographs that were shot by Scotti were subjected to a variety of lighting settings, and he worked hard to get accurate segmentation findings.

Piura et al. [8] carried out the procedure of segmenting white blood cells. A neural network was employed for the categorization, and a method called edge detection was applied for each individual leucocyte. The neural network's

ability to distinguish lymphoblasts was trained using morphological characteristics.

An automated technique that counts the number of blasts in the microscopic blood picture was suggested by [15]. In order to determine whether the picture included white blood cells, a threshold was applied to the S component of the HSV color space. The results that were obtained were really astounding; nevertheless, the issue with the system is that there is no way stated for setting the optimal threshold for the better segmentation. This presents a dilemma. It has not been possible to successfully extract the nucleus, cytoplasm, or nucleolus from the pictures of lymphocytes due to the absence of any feature extraction as well as any classifier that might be applied.

The two-stage segmentation techniques have been implemented, which has resulted in an increase in the amount of time required for computing. It is challenging to choose the optimal threshold for segmentation since the photographs were captured using a variety of lighting conditions, which makes the task more challenging overall.

[16] suggested a method for the extraction of the nucleus. This plan, which is based on the picture forest transform, has included the watershed transform as one of its components. The information on the size distribution was used in the process of extracting cytoplasm. If the cytoplasm is not in the form of a sphere, then this plan did not work out very well.

A method for the segmentation of cells was suggested by [17]. The picture is partitioned into groups using the \$Smithrils* mahram* mahram\$ color space, and the algorithm is responsible for its creation. This method performed very well for the segmentation of the nucleus; however, the cytoplasm extraction, which is just as crucial as the nucleus segmentation, has not been addressed.

By using the approach of cropping the full cell, it is not possible to see the true area of the complete cell in a clear way [18]. A fuzzy \$matric\$-Mean clustering is applied by Thereupon to divide single cell photographs of white platelets in the bone marrow into two categories: nucleus and non-nucleus. When there are more than two clusters, the processing time is exponentially longer.

The methodology known as "Feature Extraction Using Wavelet for Medical Picture Classification" was first suggested by Rajaei A and Rangarajan L. The process by which they extracted wavelet highlights from medical pictures, each of which has its place among the several modalities. When it came to the deterioration of the medical photos, they relied on Discrete Wavelet Transform. The mean and standard deviation of the deleted features were sent into the K-Nearest Classifier as a contribution for characterization, and \$99.96\%\$ more accuracy was achieved with DWT than DCT.

The technique known as "DWT Based Feature Extraction employing Edge Tracked Scale Normalization for Improved Face Acknowledgment" was developed by Rinki B.P. and his colleagues. They proposed a novel strategy for increased rate in confront acknowledgment frameworks, in which the ETSN process performed edge discovery (as a preprocessing), alongside the utilization of scale standardization to eliminate the foundation points of interest. This allowed them to improve the rate at which the frameworks could recognize faces. In this technique, the DWT is used to extract wavelet

highlights, and then the Binary Particle Swarm Optimization (BPSO) is used as an element extraction strategy [19]. They applied the Euclidean classifier as a method for the classification operation.

III. MATERIALS AND METHODS

The advancements and problems that are now being faced in the area of artificial intelligence (AI) have been focused on narrowing the decision-making gap that exists between humans and robots.

The field of artificial intelligence known as machine learning (ML) is focused on the development of computer programs that can teach themselves from data and improve themselves through experience with little to no assistance from humans or programming. Pattern recognition strategies that are based on machine learning may be classified as either supervised or unsupervised learning approaches. The supervised group includes methods such as linear regression, support vector machines (SVM), artificial neural networks (ANN), and others like them. The unsupervised category includes segmentation techniques such as clustering by k-means and fuzzy C-means (FCM), among others like them. SVM, k-Nearest Neighbours (k-NN), Naive Bayes, and Ensemble classifiers were some of the traditional machine learning-based methods that were used in this work to compare the classification performance.

A widely used machine learning (ML)-based tool for classification and regression, the support vector machine (SVM) was created by Vladimir Vapnik at AT&T Bell Labs. It utilizes a statistical learning paradigm as its foundation. With the assistance of a collection of characteristics derived from the data, a support vector machine (SVM) may map fresh instances to one of the two classes by using a set of training data in which each of the 64 examples has been assigned to one of the two classes. The support vector machine (SVM) method presents each datum as a point in an n -dimensional space, where n is the number of features, and the classification is accomplished by using a hyper-plane that divides the two classes. In order to configure the decision function, the support vector machine (SVM) has the ability to make use of a number of different kernel functions, including the liner kernel, quadratic kernel, polynomial kernel, radial basis function kernel, and multilayer perceptron kernel. In this body of work, a $K(K-1)/2$ binary support vector machine model with a linear kernel is used. K is the total number of distinct class labels (levels).

The nearest neighbors-based classifier is a basic supervised machine learning technique that is founded on the idea of proximity. It operates on the assumption that items that are physically adjacent to one another are more like one another. A new data point is always placed in the class to which its k nearest neighbours belong, and the k -NN classifier does this automatically (Fix and Hodges, 1951). While searching for the most orderly neighbours of the new sample, a straightforward distance metric such as the Euclidean distance may be used.

The Naive Bayes classifier is a probabilistic machine learning model that is part of a larger family of probabilistic classifiers that are based on the traditional Bayes' theorem. This family of probabilistic classifiers is known as the Naive Bayes group.

The core premise of Naive Bayes is that every characteristic of a sample is independent and contributes equally to the result that a given sample belongs to a class. This is known as the "independence of the features." The posterior probability that a sample belongs to each class is estimated using this procedure, and after that, the sample is allocated to the class that has the greatest posterior probability. The Naive Bayes model allows for the use of 65 different conditional distributions, including the Normal (Gaussian), Kernel, Multinomial, and Multivariate Multinomial distributions.

A. Ensemble Classifiers:

In ensemble learning a new classifier is generated from a set of base classifiers in such a way that the performance of the new classifier is superior to that of any constituent classifier. This is accomplished by ensuring that the new classifier is generated in such a way as to maximize the likelihood of success. The overall design of an ensemble classifier is seen in Fig. 1. This architecture is made up of a number of learners or hypotheses, which are then integrated to produce an ensemble of the component learners. The presence of reap hook cells as a gift in RBCs was analyzed shown in Fig. 2.

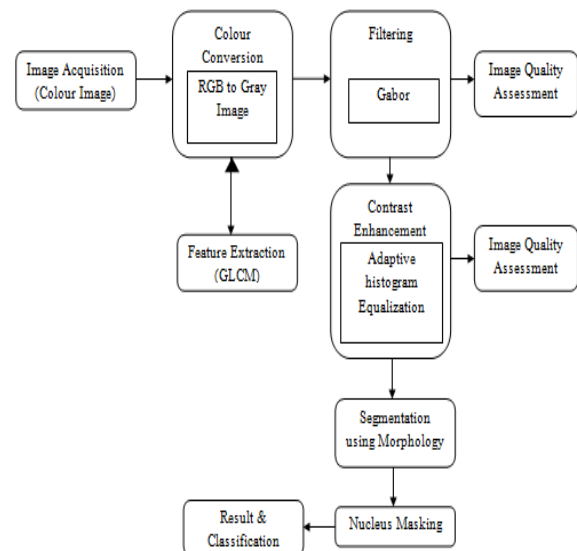


Fig. 2 MATLAB Unit

1) *Image Resizing*: Deep Learning (DL) is a subfield of machine learning, and a Deep Neural Network (DNN) is nothing more than an Artificial Neural Network (ANN) that has numerous hidden layers added to it. The inability of traditional machine learning-based algorithms to analyse natural data in its raw form is a significant limitation of these types of approaches. However, a DNN can process natural data in its raw form and it can automatically learn the features of the input data without requiring as much complex hand-engineered feature extraction as is necessary for conventional machine learning-based methods.

Combination 66 has shown outstanding performance in a variety of difficult machine learning tasks, including voice recognition, face recognition, and picture classification, among others. DNNs come in many forms, including Multilayer Perceptron's (MLP), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and

others. Only CNNs are included in this proposed work since the classifications obtained from this research are mostly determined by CNNs.

2) *Neural Networks Composed of Convolutions:* A Convolutional Neural Network, often known as a CNN or Convent, is a kind of deep neural network that has the ability to assign learnable weights and biases to certain characteristics of an input picture in order to arrive at a conclusion that differentiates one image from another. CNNs are frequently employed in deep learning, and since they are well suited for using pictures as inputs, this kind of learning makes extensive use of them. The organic configurations of a visual cortex, which comprises of both simple and complex cells, are the source of inspiration for CNNs. These cells become active according to certain subregions of the visual fields that are referred to as receptive fields. In a similar manner, the neurons in a convolutional layer of a neural network are not entirely linked to one another like the neurons in traditional neural networks, but rather they are connected to the subregions of the layer below them. Hence, the neurons are not sensitive for locations that are not included inside the image's subregions. Fig. 3 is an example of the conventional design of a CNN. Image Input Layer, Convolution Layer, Rectified Linear Unit (ReLU), Max Pooling Layer, Fully Connected Layer (FC), Soft max Layer, and Classification Layer are the typical layers that can be discovered in the majority of CNNs. Other common layers include Soft max Layer and Classification Layer. The table below provides an overview of the most essential functions of this level:

$$AR = A_2/A_1 = 0.5828 \quad (1)$$

In most cases, a Magnetic Resonance Imaging (MRI) scanner is used to do clinical analysis of osteoarthritis (OA). This scanner also provides clinical photos that may be used to study infections that are present in a human body. The use of a magnetic resonance imaging (MRI) scanner, which employs a combination of attraction and radio wave signals but does not provide ionizing radiations, provides a better view on the structure of the organs in the human body. The development of the MRI scanner has allowed for improved image handling as well as the ability to identify diseases such as knee bone and ligament segments. The procedure used by the scanner is simple and does not cause any discomfort, and it generates anatomic constructs in the form of photographs together with spatial data. The imaging mechanism of an MRI scanner is compatible with the human body, which is mostly composed of water (about 75%). An attractive magnetic field is sent to the body of the patient via the MRI scanner. By applying radiofrequency radiation to the hydrogen particle that is found in the human body, excitations of the hydrogen atoms may be brought about with extraordinarily little additional energy. As the radio frequency impulses come to an end or use up a significant amount of energy, the hydrogen particles release the energy that they have taken up, so revealing the structure of the organ. The X-ray scanner provides a variety of MR image successions, including Magnetic Resonance Spectroscopy (MRS), Fluid Attenuated Inversion Recovery (FLAIR), T1-Weighted (T1-W), and T2-Weighted (T2-W) pictures.

Different sequences of knee MR photos may assist early distinguish complex tissue structures and unique parts, which is beneficial for both radiation and clinical assessment as a whole. The Siemens-Aera 1.5 Tesla Magneto Avant Tim MRI scanner is often used these days; it provides a variety of MR views thanks to its layout of 48 channels. Knee MR cuts completed on 100 individuals with a combined total of 28 cases of osteoarthritis and scratches had a cut thickness of 0.6 millimeters.

The MR knee cuts that have a place with clinical archives, the NTA challenge-2013, and the Knee Web dataset have been utilized to support this notion. These clinical photos include sections that are representative of the femur, the tibia, and the ligaments. The magnetic resonance imaging (MRI) scanner has been put to use in the medical field to take pictures of the human body's internal organs. These pictures are helpful for determining the severity of an illness and for predicting its course. The clinical dataset was obtained from patients aged 2 to 58 who were determined to have i) Normal knee bone, ii) Osteoarthritis impacted knee, and iii) ligament area has gotten the MR knee bone pictures. The dataset was determined by using the scanner device that was mentioned earlier. The patients' ages ranged from 2 to 58. The clinical photos were analyzed using ground truth pictures that were physically portioned by human experts. This was done to assist with important decisions and supply limited ground bits of insight to many clinical datasets. This postulation makes use of the dataset provided by the NIH challenge-2013, which includes ground truth photos for the purpose of providing unambiguous acceptance. With the use of the sanctioned ground truth photos, the division precision of the suggested computation has been acclimatized in a clear and unambiguous manner. The Knee Online Simulated data collection includes several photographs of the knee, most of which are packed to the brim with Rician clamors. The photographs are put through the suggested computation for basic division at each possible limit, and the results are used to determine whether or not they should be shown. The images have been processed using a system that is outfitted with an Intel processor that has a clock speed of 2.1 GHz and 4 gigabytes of random access memory (RAM). To have unambiguous information on the suggested calculation's division greatness, this postulation made use of fifty of the most clinically tested MR cuts from the NIH, seventeen MR cuts from the clinical dataset, and 45 MR cuts from the Knee Web information base.

B. Module Description

1) *Input Image:* With the assistance of a 1.5 T MR scanner and a circularly energised communication get limit curl, a total of 2576 knee bone MR photos with a resolution of 512 by 512 pixels were obtained from 16 patients. In order to construct the three-dimensional image, the photos stored in the individual voxels must be obtained. Isotropic voxels are used to create the images, and each of those images may be recreated in one of three different planes, all of which are referred to in area 1.

The image that is often obtained from an MRI scan will typically include noise. As a consequence of this, an anisotropic channel is used in this work in order to sort through all of the disturbance. By using an anisotropic dispersion approach, it is possible to prevent the edges of the image from being covered when applying channels. This is made possible because of the method's power. This channel

is of the non-straight variety, which allows for dispersion in the area that is homogenous but restricts dispersion at the boundaries shown in Fig. 3.

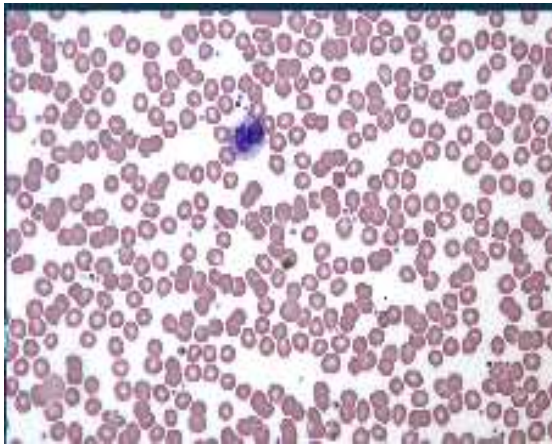


Fig. 3 Input Image

2) *Gray Image*: Moreover, the fourth and fifth convolutional layers, each with 128 and 192 filters, are applied to the down sampled pictures together with the third maximum pooling layer. This produces a feature map with dimensions of 25 by 25. The model carries on through the other levels until it reaches completely linked layers, at which point all of the neurons in each layer are connected to all of the neurons in the layer below them. In the end, the Softmax classifier is used to determine if the pictures should be considered normal, benign, or malignant. The precision of the second architecture has been increased compared to that of the first architecture. In addition, there is one completely linked layer included in architecture 3, which has the same layer descriptions and hyperparameters as the previous architectures shown in Fig. 4.

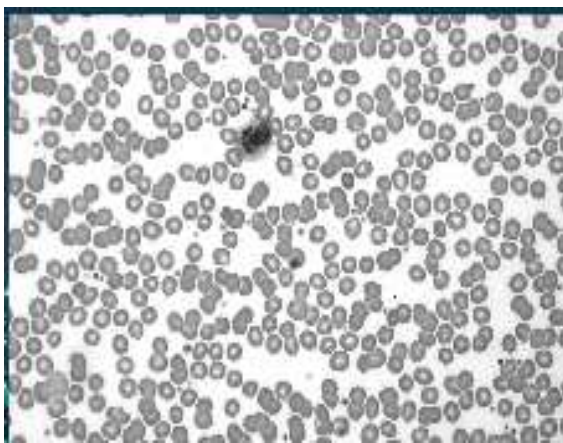


Fig. 4 Gray Image

C. Filtering

The third version of the DCNN network also has seven convolutional layers, four pooling layers, two fully connected layers, and a Softmax classifier. The input pictures have a patch size of 256 by 256, and the first convolutional layer applies 64 filters with a filter size of 5 by 5 to those images. With the use of 2 by 2 filtering, the max pooling layer is able to successfully lower the output size of the preceding

convolutional layer. This picture is then passed on to the second and third convolutional layers, where 98 and 128 filters are respectively applied to the image. The output image size of the first pooling layer is 126 by 126. After that, the max pooling technique is done, which results in an output picture size of 59 by 59 pixels shown in Fig. 5.

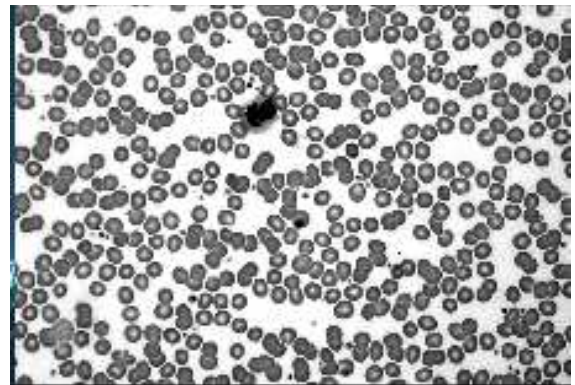


Fig. 5 Median Filter

The second iteration of the Deep Convolutional Neural Network (DCNN) has thirteen layers, including seven convolutional layers, four pooling layers, a fully connected layer, and a Softmax classifier. The size of the filters in all the convolutional layers is 5 x 5, while the size of the filters in the pooling layers is regarded to be 2 x 2. 64, 96, 128, 192, and 256 are the relative numbers of filters that may be found on feature maps that are found in convolutional layers.

D. Contrast Enhancement

The CT picture of the lung is used as an input for training and testing, both of which are retrieved via datasets provided by LIDC and Kaggle. The following metrics are used to compare and assess the performance of the proposed DCNN architecture in terms of accuracy, precision, recall, and specificity:

1) *True Positive*: Both the value of the actual class and the value of the anticipated class are accurate representations of the situation.

The term "false positive" refers to the situation in which the actual class is incorrect, but the anticipated class is accurate.

2) *True Negative*: Both the value of the actual class and the value of the predicted class are false. The value of the actual class is false in Fig. 6.

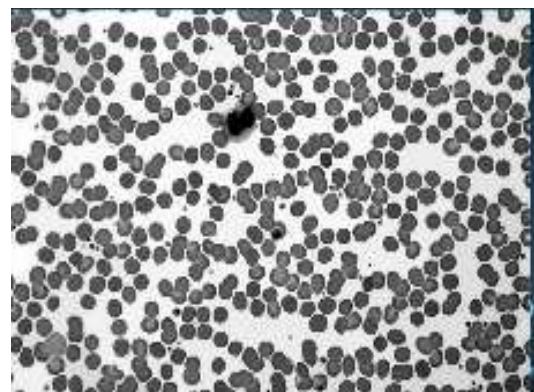


Fig. 6 Adaptive Histogram Equalization

E. GLCM

GLCM, which stands for grey level spatial dependency matrix, is a statistical approach for analyzing texture that considers the interaction that pixels have with one another spatially. Deep learning utilizes a recognition technique or algorithm, and image processing use an extraction approach that starts with a beginning set of measured data to build a feature extracted value that is meant to be informative. Dimensionality reduction is closely tied to quantitative analysis, and feature extraction is a part it shown in Fig. 7.

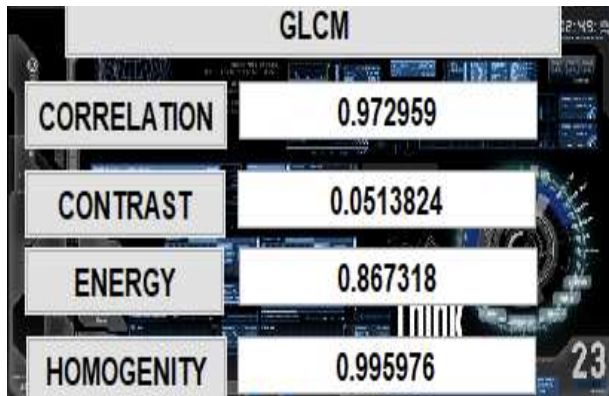


Fig. 7 Feature Extraction using GLCM.

IV. EXPERIMENTAL RESULTS

In comparison to the other two suggested architectures, the DCNN architecture 3 achieves the best results in terms of accuracy, precision, recall, and specificity.

Hence, DCNN architecture 3 is chosen for comparison with other well-known designs, such as Alex Net and GoogleNet, as well as classification techniques (SVM and ANN). revealed the Alex Net architecture for the first time after they won the ILSVRC-2012 competition (ImageNet Large Scale Visual Recognition challenge). The architecture of Alex Net is composed of three convolutional layers, five pooling layers, and three fully-connected layers, while the architecture of GoogleNet is composed of three convolutional modules, two max pooling modules, and nine inception modules. Moreover, a convolutional layer and a max pooling layer are included in each and every Inception module.

GoogleNet came out on top in the ILSVRC14 competition this shown in Fig. 8.



Fig. 8 MATLAB unit

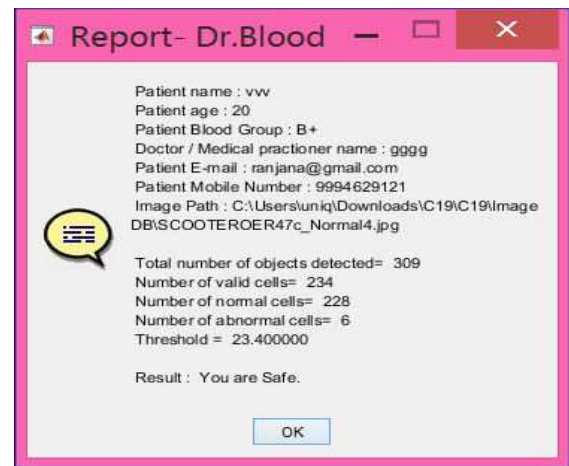


Fig. 9 Result

Now, the expanding field of medical picture categorization is the use of algorithms that are based on deep learning. Deep learning-based deep convolutional neural networks (DCNN) architectures for detection and categorization of CT lung images are discussed in this Fig. 9

The in-depth explanation of the layers of the deep learning architecture along with the three kinds of DCNN architectures that were suggested were covered here. This chapter provides information on the input and output feature sizes, as well as the hyperparameters that were employed in the suggested architectures. The CT lung picture is used as the input, and several suggested DCNN designs are used to determine if it should be considered normal, benign, or malignant. Based on the findings of the experiments, it can be concluded that the suggested architecture 3 provides superior performance in comparison to other current structures for the categorization of CT lung images. This is measured in terms of accuracy, precision, recall, and specificity in Table I. Neural Network Diagnostic results: Overall success rates

TABLE I. NEURAL NETWORK DIAGNOSTIC RESULTS: OVERALL SUCCESS RATES

Overall samples	testing	training
35/35	35/50	35/75
56%	73%	88%

TABLE II. STATISTICAL ESTIMATES FOR THE FEATURE NUCLEUS SHAPE (S)

Types of cells	Venture description			
	Min	Mean	Max	SD
Cell 1	1.1134	1.1188	1.1654	1.1228
Cell 2	1.1133	1.1152	1.1132	1.1127
Cell 3	1.1131	1.1181	1.1219	1.1153
Cell 4	1.1119	1.1165	1.1238	1.1139
Cell 5	1.1131	1.1276	1.1394	1.1197

The Table II tumor analysis using medical imaging tool is a complicated and challenging task, because computer aided tools are still far away to confirm tumor malignancy and its growing rate. During the manual analysis, the physician have to take sample tissue from lungs and which is undergone through the biopsy test to decide the type of tumor. To overcome these issues, presented the automatic image classification algorithm. The proposed work was categorized into preprocessing and classification steps. In the first step,

the filtering is applied for noise removal, then the image is segmented and enhanced since medical images affected by low contrast. This preprocessing was carried out before classification for accurate tumor classification with least time. The features like, size, shape, spreading area and growth rate of cells are used to measure the type of tumor. Then the tumor can be further classified as either benign or malignant based on the above mentioned features, if cells grow rapidly with uneven structure then it is classified as malignant otherwise benign tumor shown in Table III.

TABLE III. STATISTICAL ESTIMATES FOR THE FEATURE AREA RATIO (A)

Types of cells	Feature description			
	Max	Min	SD	Mean
Cell 1	1.1711	1.2335	1.2971	3.5215
Cell 2	1.8111	2.4587	4.4111	1.3795
Cell 3	1.1711	2.8361	4.5111	1.1178

Two investigations conducted a classification of AML into subtypes, including M3, M4, and M5, respectively, classified AML into subtypes M0 to M7 using a variety of methods, one of which was DL, with the assistance of CD (Cluster differentiation) markers. The researchers achieved an accuracy of 98.45% when separating M3, M4, and M5 utilizing DL-based methods.

The proposed research was able to successfully classify AML into the subtypes M3, M4, and M5 by using the CNN, ResNet-50, on its own in method 1, achieving an accuracy of 96.43%. In method 2, classification using the DNN features, classification by SVM and k-NN, and classification by k-NN all achieved an accuracy of 100% in Table III.

V. CONCLUSION

Orthopedic specialists have challenges in diagnosing osteoarthritis since the disease manifests itself in a manner that is almost identical across all joints. The purpose of this proposed work is to provide an original combination of soft computing methods, such as the active contour approach and the clustering methodology. In this proposed work work, the automated segmentation of MR knee bone structures is accomplished using the Douglas Richford splitting algorithm, which is a subsection of the proximal splitting technique, and the Eagle algorithm with a multi stochastic objective process. Both algorithms were developed by Eagle Scientific. The advantage of such an addition is that it compels the algorithm to visit each pixel in the cluster group and regulates the greedy character of the active contour algorithm (the deletion of a pixel during the clustering process). Both benefits are important for achieving optimal results. The convergence factor of the algorithm is controlled by the arithmetic membership function, and the splitting method is enhanced with the assistance of the firefly results. The segmentation results produced by the algorithm are reliable for the purpose of MR image analysis of the knee bone. In this body of study, the Dice Similarity Coefficient (DSC), Mean Squared Error (MSE), Region of Interest (ROI), mean, Standard Deviation (SD), and Peak Signal to Noise Ratio (PSNR) data were used in order to do evaluations on the variable in question. SD values were provided as 73%, and ROI values were supplied as 88.95%, respectively. Comparisons are made between the effectiveness of this algorithm and that of more conventional methods. It has been shown that the technology in question allows for improved vision of the knee bone cartilage

structures, which will be of tremendous assistance to orthopedic practitioners in the course of therapy.

The Douglas Richford dividing method that was proposed resulted in an increase in both the standard deviation and the return on investment.

REFERENCES

- [1] J. M. Bowman, B. Chown, M. Lewis, and J. M. Pollock, "Rh isoimmunization during pregnancy: antenatal prophylaxis." *Can. Med. Assoc. J.*, vol. 118, no. 6, pp. 623–7, Mar. 1978.
- [2] B. H. Davis, S. Olsen, N. C. Bigelow, and J. C. Chen, "Detection of fetal red cells in fetomaternal hemorrhage using a fetal hemoglobin monoclonal antibody by flow cytometry." *Transfusion*, vol. 38, no. 8, pp. 749–56, Aug. 1998.
- [3] J. Bowman and J. Pollock, "Rh-immunization during pregnancy and the grandmother theory," *J. Pediatr.*, vol. 93, no. 2, pp. 313–314, Aug. 1978.
- [4] C. Chen, S. Cheng, C. Lee, F. Chang, C. Wu, and Y. Yuh, "Fetomaternal Hemorrhage," *J. Med. Sci.*, vol. 23, no. 4, pp. 231–234, 2003.
- [5] Z. Wang, J. Shi, Y. Zhou, and C. Ruan, "Detection of red blood cell-bound immunoglobulin G by flow cytometry and its application in the diagnosis of autoimmune hemolytic anemia." *Int. J. Hematol.*, vol. 73, no. 2, pp. 188–93, Feb. 2001.
- [6] Nayak, J. P., Parameshachari, B. D., Soyjaudah, K. S., Banu, R., & Nuthan, A. C. (2017, December). Identification of PCB faults using image processing. In 2017 International Conference on Electrical, Electronics, Communication, Computer, and Optimization Techniques (ICEECCOT) (pp. 1-4). IEEE.
- [7] W. C. Warger II, J. A. Newmark, B. Zhao, C. M. Warner, and C. A. DiMarzio, "Accurate cell counts in live mouse embryos using optical quadrature and differential interference contrast microscopy," in *Pro. SPIE*, Feb. 2006, pp. 609 009–12.
- [8] D. M. Pelikan, S. a. Scherjon, W. E. Mesker, G. M. de Groot- Swings, G. G. Brouwer-Mandema, H. J. Tanke, and H. H. Kanhai, "Quantification of fetomaternal hemorrhage: a comparative study of the manual and automated microscopic Kleihauer-Betke tests and flow cytometry in clinical samples." *Am. J. Obstet. Gynecol.*, vol. 191, no. 2, pp. 551–7, Aug. 2004.
- [9] D. Huh, W. Gu, Y. Kamotani, J. B. Grotberg, and S. Takayama, "Microfluidics for flow cytometric analysis of cells and particles." *Physiol. Meas.*, vol. 26, no. 3, pp. R73–98, June 2005.
- [10] Y. Mundee, N. C. Bigelow, B. H. Davis, and J. B. Porter, "Simplified flow cytometric method for fetal hemoglobin containing red blood cells." *Cytometry*, vol. 42, no. 6, pp. 389–93, Dec. 2000.
- [11] Piyush Kumar Pareek, Chethana Sridhar, R. Kalidoss, Muhammad Aslam, Manish Maheshwari, Prashant Kumar Shukla, Stephen Jeswinde Nuagah, "IntOPMICM: Intelligent Medical Image Size Reduction Model", *Journal of Healthcare Engineering*, vol. 2022, Article ID 5171016, 11 pages, 2022. <https://doi.org/10.1155/2022/5171016>
- [12] Srinivas, M., and C. Krishna Mohan. "Classification of medical images using edge-based features and sparse representation." In 2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 912-916. IEEE, 2016.
- [13] M. Sathya, M. Jeyaselvi, Shubham Joshi, Ekta Pandey, Piyush Kumar Pareek, Sajjad Shaukat Jamal, Vinay Kumar, Henry Kwame Atiglah, "Cancer Categorization Using Genetic Algorithm to Identify Biomarker Genes", *Journal of Healthcare Engineering*, vol. 2022, Article ID 5821938, 12 pages, 2022. <https://doi.org/10.1155/2022/5821938>
- [14] Anil Audumbar Pise, Khalid K. Almusaini, Tariq Ahamed Ahanger, Ahmed Farouk, Kumud pant, Piyush Kumar Pareek, Stephen Jeswinde Nuagah, "Enabling Artificial Intelligence of Things (AIoT) Healthcare Architectures and Listing Security Issues", *Computational Intelligence and Neuroscience*, vol. 2022, Article ID 8421434, 14 pages, 2022. <https://doi.org/10.1155/2022/8421434>.
- [15] Vaghela, H. P., Modi, H., Pandya, M., & Potdar, M. B. (2015). Leukemia detection using digital image processing techniques. *Leukemia*, 10(1), 43-51.